

Ahmed Ikram | Data Engineering & Analytics

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PROFILE

3rd-year BSc Computer Science (Data Science & AI) student at the University of Dundee, on track for a First-Class degree. Specialises in event-driven pipeline engineering, cloud lakehouse architecture, and ETL/ELT orchestration - having built a production-grade Kafka ingestion system processing 2.5M+ events at 100+ msg/sec on AWS, and a serverless Bronze→Silver→Star Schema medallion pipeline on Azure Databricks DLT ingesting 153,380 rows with zero dropped across 7 quality gates, provisioned via Terraform with OIDC-federated CI/CD. Applies engineering rigour throughout: dual-dialect dbt models, 305 pytest tests plus 121 dbt data tests, 8×+ JDBC performance optimisation, and Grafana observability from day one. Seeking a Data Engineering or Analytics post-graduation role to contribute production-grade pipeline design, dbt modelling, and cloud infrastructure skills in a data-first team.

EDUCATION

High School of Dundee

Aug 2017 - June 2023

- National 5 Computer Science, Maths, English, Economics, Physics, Chemistry, Biology (A)
- Higher Computer Science, Maths and English (A), Higher Chemistry (B) and Higher Physics (C)
- Advanced Higher Computer Science (C)

University of Dundee - BSc (Hons) Computer Science (Data Science and AI)

Sep 2023 - Present

- Expected Graduation: June 2027

PROJECTS

W3C Web Logs ETL Pipeline | *Apache Airflow, Python, Databricks, dbt, Apache Spark, PySpark, Azure SQL, Delta Lake, Terraform, Docker, Power BI, DAX, Power Automate, Grafana, Prometheus* | Feb 2026 - June 2026

- Ingested 93 W3C IIS log files (155,380 rows, 0 dropped) on Azure Databricks DLT via Auto Loader with a custom parser detecting 14- vs 18-field IIS formats at parse time; Silver enriches every row via a consolidated MaxMind GeoIP struct UDF - 3.5× faster than 7 separate UDFs - achieving 99.99% known-country coverage across 153,377 enriched, 31-column rows across 7 DLT quality expectations
- Cut Silver→Azure SQL export 8-9× (413s → ~45s) via Spark-side pre-filtering before collect(), tuple(row) serialisation over asDict() (4.7M fewer dict allocations), and removing redundant .count() scans; dimension tables (geolocation & user-agent) built via SQL MERGE upsert with SHA-256 hash natural keys and -1 sentinel rows preserving FK integrity
- Built 16 dbt models (10 staging + 6 marts) compiling against both PostgreSQL and T-SQL from a single codebase via inline Jinja branches and 18 compatibility macros; incremental fact table dedupes on a 12-component MD5 key, catching 213 duplicate rows across 155K+ raw records; 121 dbt tests (not_null, unique, accepted_values, relationships, expression_is_true) enforce referential integrity across all 9 dimensions before data reaches Power BI
- Orchestrated via 4 Airflow DAGs with Dataset-driven decoupling - no ExternalTaskSensor, no hard-coded DAG IDs; provisioned 30+ Azure resources across 2 Terraform configurations with a fully OIDC-federated GitHub→Azure auth chain and zero static credentials; 613 tests across 5 tiers gate every push; 3 Grafana dashboards (23 panels) fed by a custom freshness probe with 10 combined Prometheus/Azure Monitor alert rules

ATM Log Aggregation, Analysis & Diagnostics Platform | *Python, Kafka, PostgreSQL, FastAPI, Redis, MLflow, Docker, AWS* | Mar 2026 - June 2026

- Led backend and data engineering on a 7-person Agile team for NCR Atleos; independently replaced the direct-DB generator with a full event-driven pipeline - designing the Kafka message bus, data lake schema, ML infrastructure and test suite entirely post-submission improving upon the team-delivered project
- Architected an Apache Kafka KRaft ingestion pipeline processing 2.5M+ events every day sustained at 100+ msg/sec across 7 log sources; 7 source-specific parsers with .get() safe defaults ensure schema drift never halts ingestion; malformed records route to a dead-letter table; hybrid Redis SET + 10K LRU deduplication guarantees exactly-once semantics across consumer restarts with exponential backoff retry on all failures
- Designed a 12-table PostgreSQL schema with JSONB payloads (new log sources require only a new parser - zero schema migrations), a v_unified_analysis UNION ALL view as the single ML query target, and 14 composite B-tree indexes supporting the most frequent query patterns and sustaining performance at 1M+ rows; validated pipeline correctness via 670 tests across 10 tiers, catching silent data loss under 50 concurrent writes and validating exactly-once semantics end-to-end

Haggis Species Classification & Predictive Modelling | *Python, Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook* | Nov 2025 - Dec 2025

- Ran a complete ML pipeline from EDA through feature engineering to model evaluation; engineered and validated 2 novel ratio-based features using sklearn pipelines (ColumnTransformer, RobustScaler) that ranked among the top-3 predictors across all models, whilst preventing data leakage

- Trained and compared 7 classifiers (XGBoost, Random Forest, Logistic Regression and others) achieving ~90% accuracy and tuned hyperparameters via GridSearchCV with 5-fold cross-validation
- Applied K-Means and DBSCAN clustering and built a Linear Regression model ($R^2=0.756$) and validated assumptions via residual diagnostics and standardised coefficient analysis

Project Management Platform with GitHub Integration | Python, Flask, PostgreSQL, *React*, *Socket.io*, *AWS*, *Docker*, *GitHub Actions*, *CI/CD* | Jan 2025 - Apr 2026

- Designed a 12-table PostgreSQL schema built around actual API access patterns - indexes on FK, join, and filter columns throughout; dedicated tables for audit logs (filterable by user, action, resource, and date with configurable retention and cleanup), typed saved reports (JSON summary and details fields), system settings, GitHub repository tracking, and bidirectional task-to-Issue/PR link associations enforced at the database layer
- Deployed production-like AWS data infrastructure: ECS Fargate & ALB serving the Flask API, RDS PostgreSQL in a private subnet accessible only from ECS at the network layer, and a React SPA on S3, CloudFront & ACM; rolling deploys tag each image with the commit SHA and latest for immutable rollback references at any prior build
- 1,452 automated tests at 85%/85% line coverage hard-gate every PR via GitHub Actions before any AWS step runs; OIDC federation authenticates the pipeline to AWS without static credentials stored anywhere

TECHNICAL SKILLS

Languages	Python, SQL, JavaScript, Bash, Typescript
Data Engineering	Airflow, dbt, Kafka, Spark, PySpark, Databricks (DLT, Auto-Loader), Dead-Letter Queues, Stream Processing, Medallion Structure
Data Modelling	Delta Lake / Lakehouse, Star Schema, Data Lake Design, JSONB Payloads, Dimensional Modelling, Schema-on-Read
Databases	PostgreSQL, MongoDB, MySQL, SQLite, Redis
Analytics & BI	Pandas, NumPy, Matplotlib, Seaborn, Power BI, Power Automate, DAX, Jupyter Notebook, Grafana, Prometheus
Cloud & DevOps	AWS (ECS Fargate, EC2, S3, RDS, IAM, CloudFront, SageMaker), Azure (Databricks, Azure SQL, Data Factory, ADLS), Docker, GitHub Actions, Linux, Terraform, OIDC / Workload Identity Federation
Testing	Pytest, dbt Tests, Data Quality Testing

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>

EXPERIENCE

Beaully Mini Market, Dundee, Scotland

Retail Assistant

September 2023 - Present

- Diagnosed and resolved Point-of-Sale and stock system failures rapidly to minimise service disruption

Zulus Peri Peri Grill, Dundee, Scotland

Front of House Worker

June 2021 - September 2023

- Trained 5 new staff members; managed competing priorities across dine-in, takeaway, and delivery orders under sustained pressure

TRANSFERRABLE SKILLS

Pipeline Design & Data Modelling

Design normalised schemas and ingestion architectures with reliability, schema drift tolerance, and edge cases as first-class concerns

Data Quality & Validation

Apply multi-stage validation - row count assertions, null checks, surrogate key fallbacks - to ensure no records are silently dropped or corrupted downstream

Root Cause Analysis

Systematically isolate pipeline failures and data issues; experienced diagnosing silent data loss, concurrency bugs, and API rate-limit failures under time pressure

Stakeholder Communication

Translate pipeline outputs and analytical findings into plain business language for non-technical audiences and project stakeholders